

Recurrence Quantification Analysis

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May 2026

1 Introduction

Traffic systems are dynamic processes that change over time and often show repeated temporal patterns, such as morning and evening rush-hour peaks. Recurrence Quantification Analysis (RQA) is useful in this context because it examines how often a system returns to similar states over time. Wallot and Leonardi (2018) describe recurrence based methods as suitable for analysing time series with complex, non stationary, or time dependent dynamics, and they present multidimensional RQA as a method for studying the temporal structure of multivariate systems.

In this project, RQA and multidimensional RQA are applied to traffic volume data. In the first part of the project, we use standard RQA to analyse one selected traffic site in detail and identify its recurring temporal structure. In the second part, we use multidimensional RQA to analyse three sites together, which allows us to examine whether nearby intersections show shared patterns of regularity, stability, and repeated traffic behaviour.

2 Data Exploration

This dataset originates from the traffic signal system, published via the Victoria (Australia) open transport data platform, and contains traffic volumes recorded every 15 minutes throughout March 2026 for multiple intersections. For this group project, we use the combined approach, where detector values are aggregated into a single total traffic volume per site at each timestamp. Detectors are individual sensors at an intersection, typically assigned to specific lanes or traffic movements, that count how many vehicles pass during each 15 minute interval. By summing these detector readings, we obtain a clean and consistent site level time series for analyzing temporal patterns and comparing traffic flow across sites.

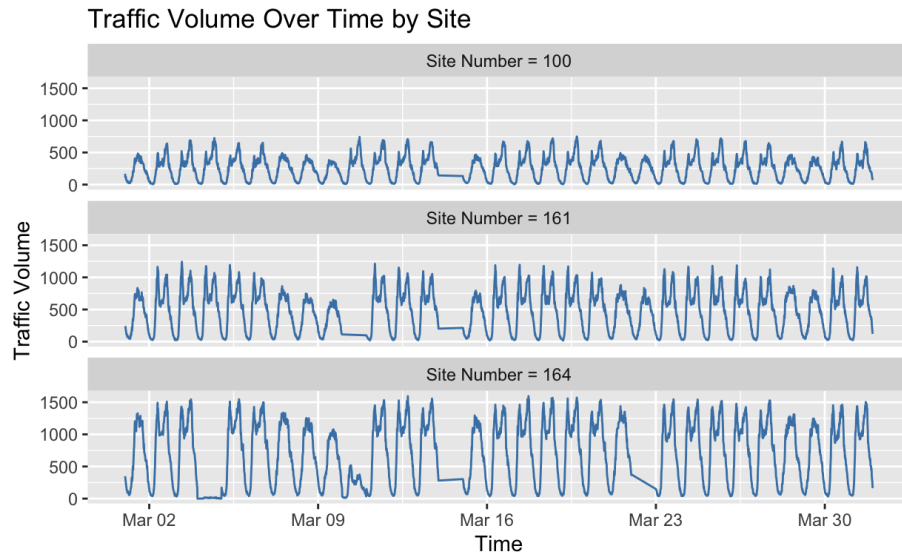


Figure 1: Traffic volume over time by site.

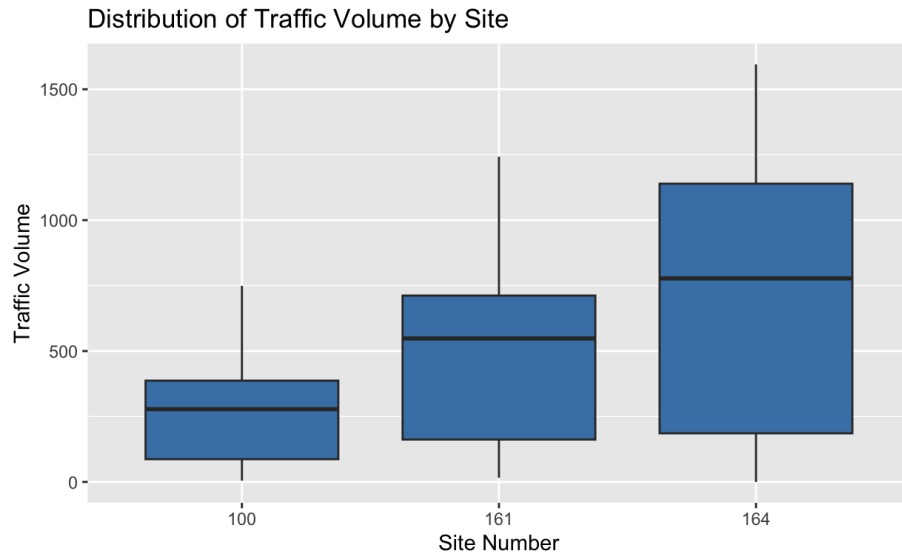


Figure 2: Distribution of traffic volume by site.

Although the full dataset contains traffic volume recordings across five intersections, this module focuses on a subset of three sites, sites 100, 161, and 164, to demonstrate the application of Recurrence Quantification Analysis.

3 RQA

3.1 RQA - Single Site Analysis

Based on the data exploration, site 164 was selected for further analysis. This site exhibits the highest variability, with an interquartile range from 185.8 to 1139.2, as well as the highest median traffic volume (Mdn = 777.5), compared to site 161 (Mdn = 548.0) and site 100 (Mdn = 278.0). Furthermore, the observed temporal data exhibit a regularity, which is expected to result in meaningful recurrence patterns.

To estimate the delay parameter τ for RQA, Average Mutual Information (AMI) was computed. A maximum lag of 96 was chosen because each lag unit corresponds to one 15 minute interval, meaning 96 lags represents exactly one full day ($96 \times 15 = 1440$ minutes). The first local minimum occurred at lag 29, corresponding to approximately 7.25 hours. This delay was selected because it captures the separation between the main recurring traffic states, especially the morning and evening peaks.

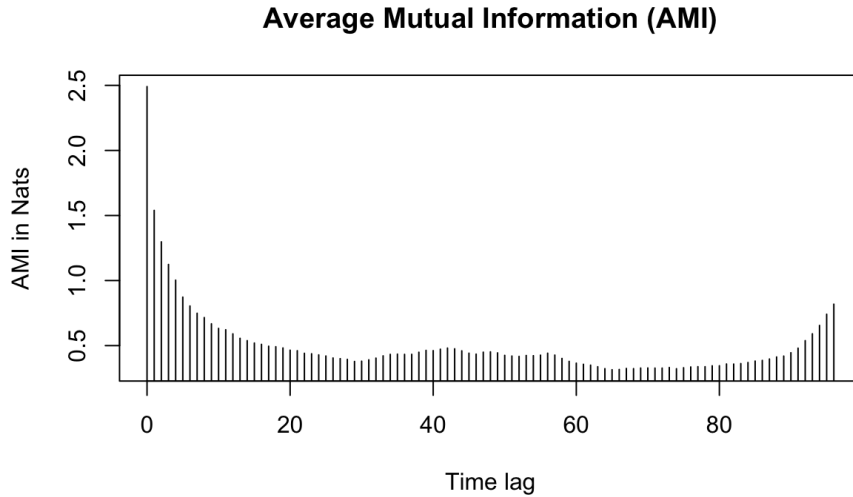


Figure 3: Average Mutual Information for site 164.

The embedding dimension, M , was estimated using the False Nearest Neighbors (FNN) method. This method identifies how many delayed versions of the time series are needed to unfold the underlying system dynamics. The FNN fraction was calculated for embedding dimensions from 1 to 20, using the previously selected delay of $\tau = 29$. The results showed a steep decrease from $M = 1$ (FNN = 44.0%) to $M = 2$ (FNN = 13.7%), followed by a slower decline at higher dimensions. After $M = 5$, where the FNN fraction reached 3.3%, further

increases in embedding dimension produced only small improvements. Therefore, $M = 5$ was selected as a embedding dimension because it captures the main dynamics of the traffic system without adding unnecessary complexity.

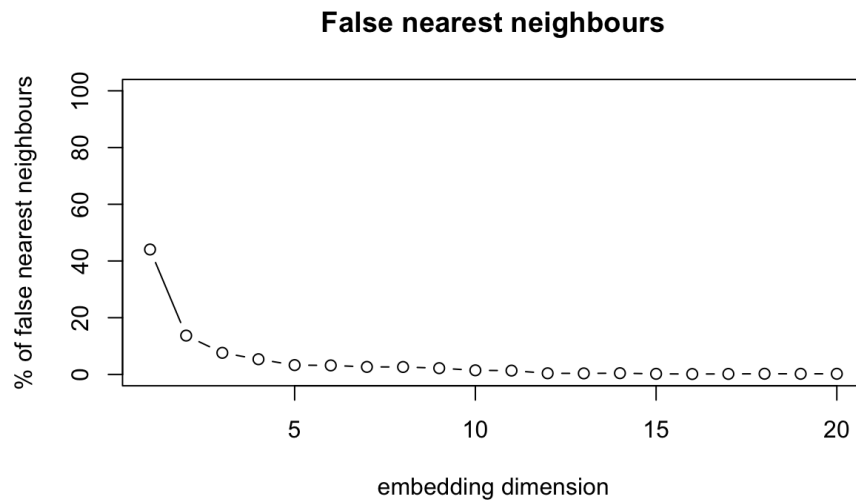


Figure 4: False nearest neighbors for site 164.

The radius was selected by testing values from 0.10 to 2.00 and inspecting the resulting recurrence rates. Following the lecture guideline that continuous behavioral signals should have a recurrence rate between 2% and 5%, radius = 0.55 was chosen. This produced a recurrence rate of 2.02%, indicating a structured and regular traffic pattern.

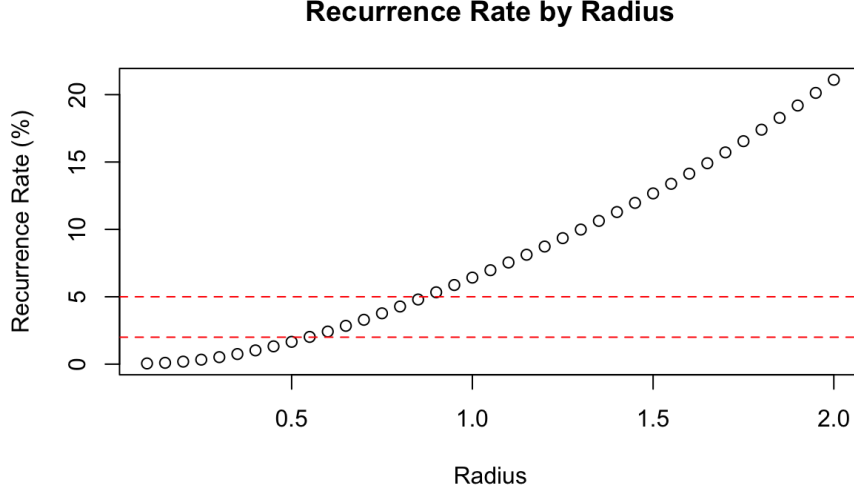


Figure 5: Radius selection and recurrence rate for site 164.

Prior to conducting the Recurrence Quantification Analysis (RQA), the time series of site 164 was standardized using z-score scaling. Consequently, the normalization and rescaling parameters within the `crqa()` function were both set to 0 (`normalize = 0`, `rescale = 0`), as the data had already been placed on a comparable scale.

For the final RQA, the previously obtained parameters were used: $\tau = 29$, $M = 5$, and $\text{radius} = 0.55$. The minimum diagonal line length and minimum vertical line length were both set to 2, as this is the minimum threshold required for a recurrent structure to be considered meaningful. The Theiler window was set to 29, matching the selected delay, to exclude points that were too close together in time and prevent trivial autocorrelation from being interpreted as meaningful recurrence. The analysis was conducted with `side = "both"`, meaning that recurrence structures on both sides of the main diagonal were included.

The RQA results indicated a recurrence rate of 1.79%, determinism of 85.89%, a mean diagonal line length of 5.55, laminarity of 92.22%, a trapping time of 4.15, and entropy of 2.09. The recurrence rate was slightly lower than the 2.02% obtained during radius selection, which is expected because the Theiler window ($\text{tw} = 29$) was applied in the final analysis to exclude recurrent points located within a short temporal interval of one another, thereby removing trivial auto-correlated recurrences from the calculation.

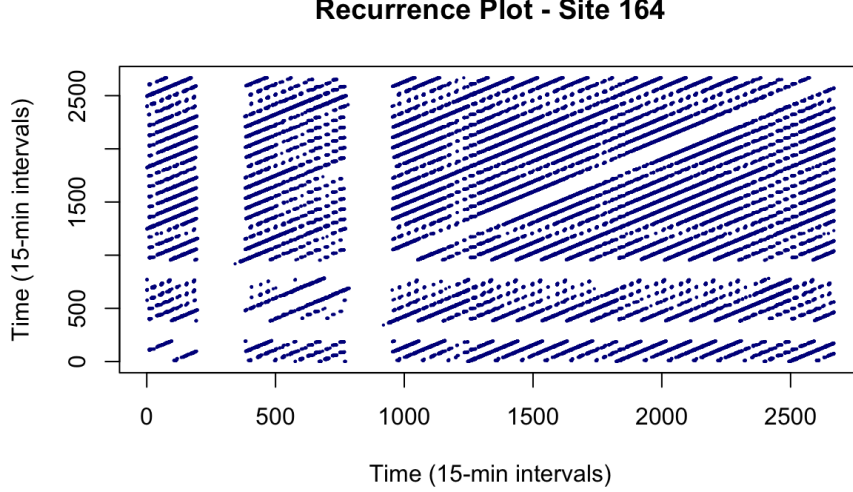


Figure 6: Recurrence plot for site 164.

The high determinism value ($DET = 85.89\%$) indicates that most recurrent points formed diagonal line structures, suggesting that traffic states often reoccurred in structured temporal sequences. The mean diagonal line length ($L = 5.55$) suggests that these recurring sequences persisted for an average of approximately 83 minutes (5.55×15 minutes) before diverging. Laminarity was high ($LAM = 92.22\%$), even higher than the determinism value, indicating that the system often remained in stable or slowly changing traffic states for consecutive observations. This is further supported by the trapping time ($TT = 4.15$), corresponding to approximately 62 minutes of sustained state stability on average. The entropy value ($ENTR = 2.09$) indicates some variation in the lengths of recurring diagonal structures, suggesting that although traffic patterns were highly structured, they were not completely uniform. Taken together, these findings suggest that traffic at site 164 showed strong temporal structure and stability, consistent with repeated daily traffic patterns rather than random fluctuations.

3.2 mdRQA - Multivariate Traffic System Analysis

After analyzing site 164 as a single time series, the next step was to examine whether the selected intersections show recurrent patterns as one multivariate traffic system. For this purpose, multidimensional recurrence quantification analysis (mdRQA) was applied to sites 100, 161, and 164. mdRQA is appropriate here because each site can be treated as one dimension of the same traffic system. Instead of checking whether one individual site returns to similar states over time, mdRQA checks whether the combined three site traffic state returns

to similar multivariate configurations.

The three site level time series were aligned by timestamp so that each row represented the same 15 minute interval across all sites.

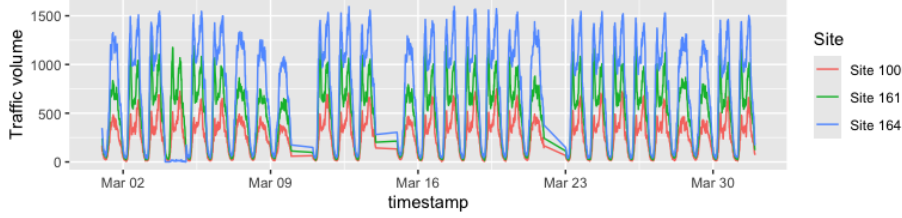


Figure 7: Multisite traffic volume over time.

The values were then standardized using z-score scaling. This was necessary because the sites had different traffic volume ranges, and without scaling, the site with the largest traffic volumes could dominate the distance calculation.

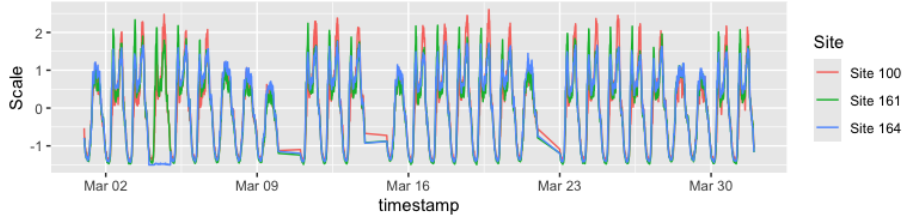


Figure 8: Standardized multisite traffic volume over time.

Before running mdRQA, the delay parameter was estimated across all three sites using a maximum lag of 96, representing one full day. The selected delay was $\tau = 21$, meaning that delayed copies were separated by 21 time steps, or 5.25 hours. This was close to the delay found in the single site RQA and therefore supported the presence of shared daily traffic structure.

After choosing the delay, the embedding dimension was estimated using the mdFnn() function. The percentage of false nearest neighbors dropped from 0.6% at dimension 1 to 0.0% at dimension 2, so an embedding dimension of 2 was selected for the mdRQA analysis.

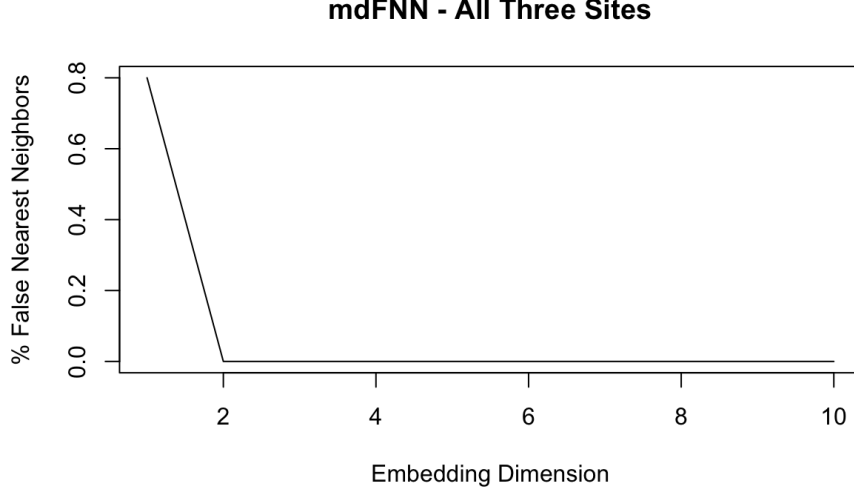


Figure 9: False nearest neighbors for mdRQA across all selected sites.

The radius was optimized for the multidimensional system using a target recurrence range of 2% to 5%. The selected radius was $r=0.549$. The same optimization procedure also returned an embedding dimension of 2 and a delay of 16, but the main purpose of this step was to obtain a suitable radius for the final mdRQA analysis. Although `optimizeParam()` returned a delay of 16, the delay of $\tau = 21$ obtained from `mdDelay()` was retained for the final analysis. This decision was made because `mdDelay()` is specifically designed to estimate the delay for multidimensional systems.

The mdRQA produced a recurrence rate (RR) of 3.52%. This means that 3.52% of all possible state comparisons were classified as recurrent. Since this value lies within the target range of 2% to 5%, the selected radius was appropriate for the analysis. The determinism (DET) was 83.24%, meaning that most recurrent points formed diagonal line structures. This suggests that the traffic system showed repeated temporal sequences rather than only isolated recurring states.

The average diagonal line length (L) was 5.02, meaning that when repeated sequences occurred, they lasted for about five time steps on average. Since each time step represents 15 minutes, this corresponds to roughly 75 minutes. The laminarity (LAM) was 90.43%, meaning that a high proportion of recurrent points formed vertical line structures. This suggests that the system often stayed in similar states for short periods rather than changing immediately. The trapping time (TT) was 4.42, meaning that these stable or slowly changing periods lasted for about 4.4 time steps on average, or roughly 66 minutes. Finally, entropy (ENTR) was 2.09, which indicates that the diagonal line lengths were moderately varied. In other words, the repeated traffic patterns were not all the

same length, but they also were not completely random.

The recurrence plot shows clear diagonal line structures across all sites, which means that similar traffic patterns repeat over time. The repeated diagonal bands suggest strong daily regularity in the multidimensional traffic system.

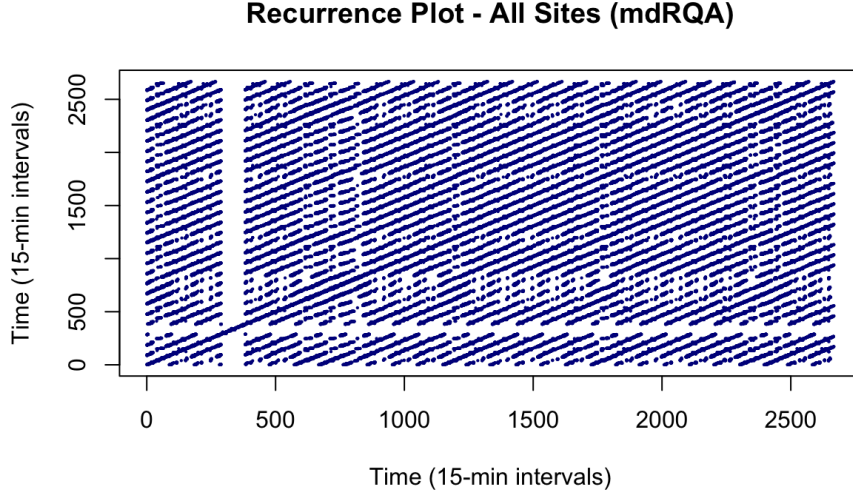


Figure 10: Recurrence plot for all selected sites using mdRQA.

Overall, the mdRQA results suggest that the three traffic sites showed a moderate amount of recurrence, with strong deterministic and laminar structure. This means that the multidimensional traffic system often returned to similar states and that these recurring states tended to form structured temporal patterns rather than appearing randomly.

4 Conclusion

This entry applied Recurrence Quantification Analysis (RQA) and Multidimensional Recurrence Quantification Analysis (mdRQA) to traffic volume data recorded at 15 minute intervals across three intersections in Victoria, Australia during March 2026, with the aim of characterizing the temporal structure and dynamics of urban traffic flow.

The single site RQA of site 164 revealed that traffic volume at this intersection is highly structured and deterministic. The recurrence rate of 1.79%, combined with a high determinism of 85.89% and laminarity of 92.22%, indicates that the system repeatedly returned to similar traffic states and that these states tended to persist over time rather than changing suddenly. The mean line length of 5.55 and trapping time of 4.15 show that recurring traffic

patterns lasted about 83 and 62 minutes. This fits the gradual rise and fall of rush hour traffic. The entropy value of 2.09 shows that the patterns were regular, but not identical each time.

The mdRQA extended this analysis to the three site system, treating sites 100, 161, and 164 as dimensions of a single multivariate traffic system. The results were largely consistent with the single site findings, with a recurrence rate of 3.52%, determinism of 83.24%, and laminarity of 90.43%. This suggests that the traffic state across all three intersections also exhibited strong temporal regularity, returning to similar multivariate configurations on a daily basis.

Together, both analyses lead to the same conclusion. Traffic flow in this area is strongly deterministic and laminar. The repeated diagonal bands in both recurrence plots support this daily rhythm. Overall, the results show that RQA and mdRQA are useful for revealing temporal patterns in traffic data and could be used in future work to detect disruptions, compare different periods, or support adaptive traffic control.

5 References

Traffic Signal Volume data - Traffic Signal Volume Data March 2026 - Open Data - Transport Victoria. (n.d.). <https://opendata.transport.vic.gov.au/dataset/traffic-signal-volume-data/resource/c34d45d2-c7b0-4cc2-be39-2d5fdbaed89a>
Wallot, S., & Leonardi, G. (2018). Analyzing multivariate dynamics using Cross-Recurrence Quantification Analysis (CRQA), Diagonal-Cross-Recurrence Profiles (DCRP), and Multidimensional Recurrence Quantification Analysis (MDRQA) - a tutorial in R. *Frontiers in Psychology*, 9, 2232. <https://doi.org/10.3389/fpsyg.2018.02232>